

EXPERIMENT 8: FOURIER RECONSTRUCTION OF MRI IMAGES

AIM:

To perform fourier reconstruction of MRI images using python in google colab.

MATERIALS REQUIRED:

Google colab software (python)

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# Load image (simulate MRI)
image = plt.imread('/content/BRAIN MRI.jpeg')

# Generate k-space
kspace = np.fft.fft2(image)

# Fourier reconstruction
kspace_shifted = np.fft.ifftshift(kspace)
reconstructed = np.fft.ifft2(kspace_shifted)
reconstructed_image = np.abs(reconstructed)

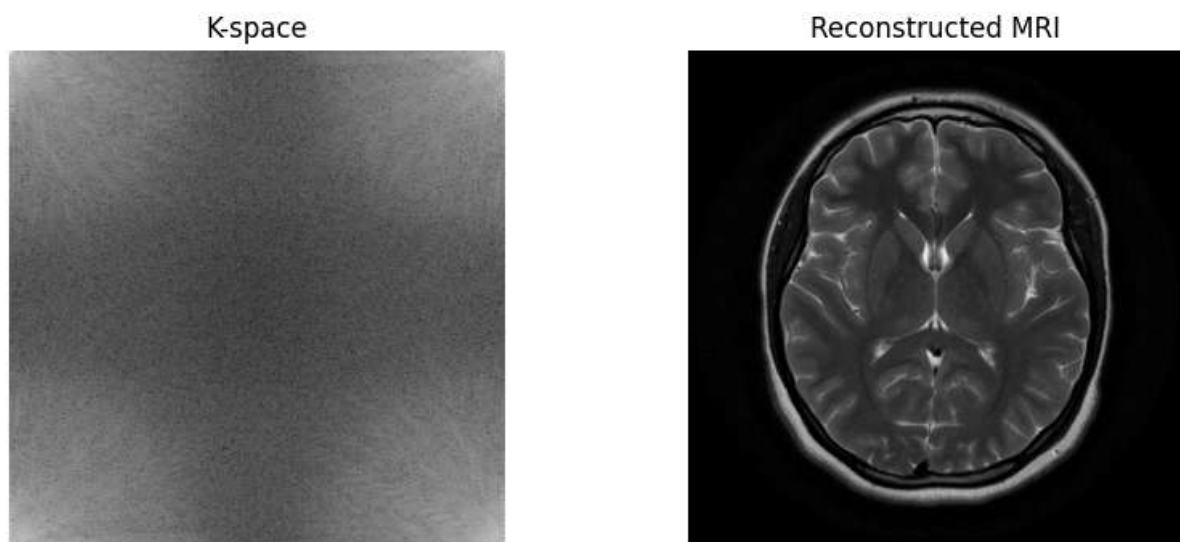
# Display
plt.figure(figsize=(10,4))

plt.subplot(121)
plt.title("K-space")
plt.imshow(np.log(np.abs(kspace)+1), cmap='gray')
plt.axis('off')
```

```
plt.subplot(122)
plt.title("Reconstructed MRI")
plt.imshow(reconstructed_image, cmap='gray')
plt.axis('off')

plt.show()
```

OUTPUT:



RESULT:

Thus, the fourier reconstruction of MRI images were performed by google colab.